

PARAS SHARMA

Ph.D. Student

School of Computing and Information,
Learning Research & Development Center,
University of Pittsburgh

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EDUCATION

Ph.D. in Computer Science

University of Pittsburgh
Advisor: Dr. Erin Walker

Aug. 2022 – Present

GPA: 3.92/4

Master of Science in Computer Science

University of Southern California

Aug. 2019 – May 2021

GPA: 3.86/4

Bachelor of Technology (with Hons.) in Computer Science & Engineering

Indian Institute of Technology (ISM), Dhanbad

Jul. 2013 – May 2017

GPA: 9.55/10

RESEARCH INTERESTS

Educational Technology; Multimodal Learning Analytics; Dialogue Systems; Collaborative Learning;
Human-Computer Interaction; Large Language Models; Learner Modeling and Adaptation

RESEARCH EXPERIENCE

Graduate Student Researcher, Learning Research & Development Center

University of Pittsburgh

Aug. 2022 – Present

Pittsburgh, PA

- Using multimodal LLMs to build theory-informed and responsive dialogue systems for learner reflections.
- Developing HMMs to understand knowledge construction processes in human-human-robot learning interactions.
- Understanding lexical alignment trajectories in teachable human-human-robot interactions
- Implementing structure and surface gestures in a teachable robotic agent to study the impact of gestures in math learning.
- Working on building a Simulated Student System to emulate learner interactions in open-ended learning environments.
- Used HMMs and Sequence Mapping to model learner activity data and extract SRL and cognitive states that learners transition through in open-design settings.
- Designed a multimodal sensing system to detect learners' sensors, programming, and dialogue activities while building a robot.

Research Intern, Institute for Creative Technologies

University of Southern California

March 2021 – Aug. 2021

Los Angeles, CA

- Developed and tested different question-answering classification techniques on the Covid-Answers dataset for building a virtual mentor.
- Used Transformers to model the similarity between input questions and answers for the mentor.

Graduate Research Assistant, Social Media Analytics Lab

Keck School of Medicine, USC

Feb. 2021 – May 2021

Los Angeles, CA

- Developed framework to fetch and analyze video advertisement data of tobacco products on YouTube and TikTok.
- Created and analyzed various metrics for the impact of tobacco product advertisements by various companies using collected video data.

Graduate Research Assistant, Viterbi iPodia

University of Southern California

Feb. 2020 – Dec. 2020

Los Angeles, CA

- Developed diverse grouping algorithm to increase cultural and intellectual diversity among cohorts.
- Designed and implemented the backend of the framework, returning groups of students.
- Collected video and transcript data for student interactions grouped using the above algorithm.

TEACHING EXPERIENCE

Teaching Fellow, School of Computing & Information

University of Pittsburgh

Aug. 2024 – Dec. 2024

Pittsburgh, PA

- Worked as the instructor for CS0011 Introduction to Computing for Scientists.
- Developed and implemented lesson plans, projects, and exams.

INDUSTRY EXPERIENCE

Machine Learning Intern, Applied Agentic Research

Meta

May 2026 – Aug. 2026

Menlo Park, CA

- Developed a user simulator for multi-turn benchmarking of Large Language Models.

Software Development Engineer, EC2

Amazon Web Services

June 2021 – July 2022

Seattle, WA

- Developed an SDK to ease the onboarding of various workloads in Launch Wizard Service.
- Integrated Launch Wizard with Alpine service to better monitor the impact of service and API failures on customers.
- Improved tagging of Launch Wizard deployed workloads to enhance the revenue generation calculations.

Software Development Engineer Intern, EC2

Amazon Web Services

May 2020 – Aug. 2020

Seattle, WA

- Designed, developed, and deployed a dedicated host tenancy feature for SQL HA in the AWS Launch Wizard Service.
- Enabled Host Resource Group Integration for the AWS Launch Wizard Service.

Senior Software Engineer, Intelligence & IoT Division

Samsung R&D Institute

June 2017 – July 2019

Bangalore, India

- Implemented reinforcement learning in a personalized Social Robot using user-interaction feedback.
- Created the Context Engine Module and reinforcement learning module of Social Robot.
- Developed and deployed Device Identity and Locksmith Microservice on AWS EC2 with data stored in DynamoDB enabling IoT devices' access to the IoT cloud utilized by partners Vodafone and Sercomm.
- Designed and implemented Android as a Thing SDK to easily onboard any Android device onto the IoT cloud and expose its functionalities as controllable OCF resources.

Software Engineer Intern, Intelligence & IoT Division

Samsung R&D Institute

May 2016 – July 2016

Bangalore, India

- Developed anomaly Sensing and recommendation Engine for Smart IoT using IoTivity.
- Built and trained a neural network model for anomaly detection using the Samsung IoT device data.

- [1] **Sharma, Paras et al.** “That Makes Me Feel Like a Helpful Robot”: Designing Hybrid Multimodal Dialogue Agents for Co-Creative Reflection. (In Review)
- [2] Angela E. B. Stewart, **Paras Sharma**, Chun Li, Ye Deng, Omotayo Madein, YuePing Sha, Janet Shufor Bih Epse Fofang, Christina Kundrak, and Erin Walker. Design Tensions for Generative AI in Education for Early to Mid-adolescent Youth: An Exploration of Autonomy, Critical Reflection, and Psychological Safety. In: Blanchard, E.G., Chen, G., Chi, M., Isotani, S. (eds) *Artificial Intelligence in Education. AIED 2026. Lecture Notes in Computer Science()*, vol 16586. Springer, Cham. https://doi.org/10.1007/978-3-032-29773-0_35
- [3] **Paras Sharma**, YuePing Sha, Janet Shufor Bih Epse Fofang, Brayden Yan, Jess A. Turner, Nicole Balay, Hubert O. Asare, Angela E.B. Stewart, and Erin Walker. 2026. Hybrid LLM-Embedded Dialogue Agents for Learner Reflection: Designing Responsive and Theory-Driven Interactions. In *Proceedings of the 2026 CHI Conference on Human Factors in Computing Systems (CHI'26)*, April 13–17, 2026, Barcelona, Spain. ACM, New York, NY, USA, 20 pages. <https://doi.org/10.1145/3772318.3791582>
- [4] **Sharma, P.** et al. (2025). Beyond Static Measures: Temporal Analysis of Lexical Alignment in Human-Human Learning With a Teachable Robot. In: Cristea, A.I., Walker, E., Lu, Y., Santos, O.C., Isotani, S. (eds) *Artificial Intelligence in Education. AIED 2025. Lecture Notes in Computer Science()*, vol 15880. Springer, Cham. https://doi.org/10.1007/978-3-031-98459-4_20
- [5] Asano, Y., Litman, D., **Sharma, P.**, Fritsch, D., King-Shepard, Q., Nokes-Malach, T., . . . Walker, E. (2025). Multi-party Lexical Alignment in Collaborative Learning with a Teachable Robot. In A. I. Cristea, E. Walker, Y. Lu, O. C. Santos, & S. Isotani (Eds), *Artificial Intelligence in Education* (pp. 123–131). Cham: Springer Nature Switzerland.
- [6] Stewart, A. E. B., Solyst, J., Bao, X., **Sharma, P.**, Buddemeyer, A., Nkrumah, T., . . . Walker, E. (2025). Who’s Got the Power? Data Feminism as a Lens for Designing AIED Engagement Systems. In A. I. Cristea, E. Walker, Y. Lu, O. C. Santos, & S. Isotani (Eds), *Artificial Intelligence in Education* (pp. 233–247). Cham: Springer Nature Switzerland.
- [7] Yuya Asano, Sabit Hassan, **Paras Sharma**, Anthony B. Sicilia, Katherine Atwell, Diane Litman, and Malihe Alikhani. 2025. Contextual ASR Error Handling with LLMs Augmentation for Goal-Oriented Conversational AI. In *Proceedings of the 31st International Conference on Computational Linguistics: Industry Track*, pages 374–386, Abu Dhabi, UAE. Association for Computational Linguistics. <https://aclanthology.org/2025.coling-industry.32/>
- [8] **Sharma, P.**, Bella, V., E. B. Stewart, A., Walker, E. (2024). Multimodal Sensing of Goals and Activities During Interactions with a Co-created Robot. In: Ferreira Mello, R., Rummel, N., Jivet, I., Pishtari, G., Ruipérez Valiente, J.A. (eds) *Technology Enhanced Learning for Inclusive and Equitable Quality Education. EC-TEL 2024. Lecture Notes in Computer Science*, vol 15160. Springer, Cham. https://doi.org/10.1007/978-3-031-72312-4_22
- [9] **Paras Sharma**, Angela E.B. Stewart, Qichang Li, Krit Ravichander, & Erin Walker. (2024). Building Learner Activity Models From Log Data Using Sequence Mapping and Hidden Markov Models. *Proceedings of the 17th International Conference on Educational Data Mining*, 584–593. <https://doi.org/10.5281/zenodo.12729890>
- [10] **Paras Sharma**, & Qichang Li. (2024). Designing Simulated Students to Emulate Learner Activity Data in an Open-Ended Learning Environment. *Proceedings of the 17th International Conference on Educational Data Mining*, 986–989. <https://doi.org/10.5281/zenodo.12730023>
- [11] Anthony Sicilia, Yuya Asano, Katherine Atwell, Qi Cheng, Dipunj Gupta, Sabit Hassan, Mert Inan, Jennifer Nwogu, **Paras Sharma**, Malihe Alikhani. “ISABEL: An Inclusive and Collaborative Task-Oriented Dialogue System”. In *Alexa Prize TaskBot Challenge 2 Proceedings*, 2023. <https://www.amazon.science/alex-prize/proceedings/isabel-an-inclusive-and-collaborative-task-oriented-dialogue-system>

HONORS AND AWARDS

3rd Position, Alexa Prize TaskBot Challenge 2. Amazon. \$50,000. 2023

Orrin E. and Margaret M. Taulbee Graduate Award for Excellence in Computer Science. Department of Computer Science, School of Computing and Information, University of Pittsburgh. 2025.

Pre-Doctoral Fellowship. School of Computing and Information, University of Pittsburgh. 2022-2023.

Viterbi Graduate Student Scholarship. Viterbi School of Engineering, University of Southern California. \$7,500. 2019.

Director's Award for Academic Excellence. Indian Institute of Technology (Indian School of Mines), Dhanbad. 2015.

Travel Scholarship. 17th Educational Data Mining Conference (EDM2024), Atlanta, Georgia. \$2600. 2024

Travel Scholarship. 25th International Conference on Artificial Intelligence in Education (AIED 2024), Recife, Brazil. \$1200. 2024

Travel Scholarship. 26th International Conference on Artificial Intelligence in Education (AIED 2025), Palermo, Italy. \$1350. 2025

MENTORSHIP EXPERIENCE

Daniel Fritsch. Sep 2024 - Present, BS Computer Science, University of Pittsburgh

Project: Determining Lexical Alignment Trajectories and Knowledge Construction Processes in a teachable human-human-robot setting.

Michael Puthumana. Sep 2024 - Present, BS Computer Science, University of Pittsburgh

Project: Hardware modifications to support external speaker and microphone sensors in the Social programmable robotic kits.

Krit Ravichander. Jan 2024 - Aug 2025, BS Computer Science and Psychology, University of Pittsburgh

Project: Building structure and surface gestures for the NAO robot.

Brayden Yan. Jan 2025 - Aug 2025, BS Computer Science, University of Pittsburgh

Project: Updating dialogue systems with LLMs to promote reflections in interactions between learners and their robot designs.

Travis Mindel. Jan 2025 - Apr 2025, BS Information Science and Psychology, University of Pittsburgh

Project: Using Media Pipe to build automated gesture replication in a NAO robot.

Jude Gilligan. Jan 2025 - Apr 2025, BS Computer Science, University of Pittsburgh

Project: Integrating real-time Lexical Alignment into the teachable robot interface.

Siyena Gezahegn. Sep 2024 - Apr 2025, BS Computer Science, University of Pittsburgh

Project: Hardware modifications to support external speaker and microphone sensors in the Social programmable robotic kits.

Zeana El Hajomar. Sep 2024 - Jan 2025, BS Computer Science, University of Pittsburgh

Project: Interface and UX Improvements for Teachable Robot.

Audrey Flood. May 2024 - July 2024, FACET Lab Summer Intern, University of Pittsburgh

Project: UI/UX design for the frontend interface of the social Programmable Robots project.

Qichang Li. Jan 2024 - Apr 2024, MS Computer Science, University of Pittsburgh.

Project: Development of a Dialogue Agent by Distilling Knowledge from a Large Language Model for Robot Programming Assistance.

Raquel Buege. Jan 2024 - Apr 2024, BS Computer Science, University of Pittsburgh

Project: Frontend implementation of the social programmable robots interface.

Veronica Bella. Jan 2023 - May 2024, BS Electrical Engineering, University of Pittsburgh

Project: Custom hardware extensions to the hummingbird robotics kit to sense immediate addition/removal of hardware components.

UNIVERSITY SERVICE

LRDC Executive Committee. University of Pittsburgh. Sep 2024 - Present

Computer Science Graduate Student Organization (President). University of Pittsburgh. Sep 2024 - Present

VOLUNTEER SERVICE

Assistant Technical Program Chair. ACM CHI Conference on Human Factors in Computing Systems (CHI 2027)

Student Volunteer. 26th International Conference on Artificial Intelligence in Education (AIED 2025)

Student Volunteer. 25th International Conference on Artificial Intelligence in Education (AIED 2024)

REVIEWING SERVICE

27th International Conference on Artificial Intelligence in Education (AIED 2026)

International Conference of Computer-Supported Collaborative Learning (CSCL 2026)

International Conference of the Learning Sciences (ICLS 2026)

TECHNICAL SKILLS

Research: Prototyping, Quantitative and Qualitative methods, Statistical Testing, Experimental Design, user modeling, Data Mining

Languages: Python, Java, C/C++

Libraries: PyTorch, TensorFlow, NumPy, pandas, Matplotlib, NLTK

Frameworks/Tools: Frontend and Backend Development, Git, Android Development, AWS EC2, Elastic Search, IBM Watson, AWS S3